

Imagine-Then-Act Chunk Selection for Lightweight Vision-Language-Action Policies

Author Names Omitted for Anonymous Review. Paper-ID [add your ID here]

Abstract—TODO.

I. INTRODUCTION

TODO.

II. RELATED WORK

A. Visual Foresight and Imagination-Based Planning

TODO cite visual MPC, visual foresight, action-conditioned video prediction, and world-model planning. Connect them to this paper’s first proof setting: using simulator clone rollouts and renders as oracle imagined outcomes before training a deployable action-conditioned visual dynamics model.

B. Candidate Trajectory Selection

TODO cite best-of-N sampling, CEM, model-predictive control, and trajectory ranking methods. Frame this paper as chunk-level candidate selection for a fixed lightweight VLA, not as post-hoc recovery after a failed rollout.

C. VLM Critics and Robotic Verification

TODO cite VLM-as-reward, VLM-as-critic, and semantic verifier work in robotics. State the boundary clearly: the VLM judge scores predicted outcome images for pre-execution selection only; final task success remains the environment benchmark signal.

D. Efficient Vision-Language-Action Policies

TODO cite SmolVLA [7], TinyVLA [8], OpenVLA [4], and broader robot foundation policies such as RT-1 [2] and RT-2 [10].

E. Agentic Planning and Feedback

TODO cite SayCan [1], Inner Monologue [3], Code as Policies [5], ReAct [9], and Reflexion [6]. Tie this to SemRob topics: semantic goal interpretation and policy steering. Keep this subsection short: the paper is not mainly an LLM-agent paper.

III. METHOD

A. Base Policy and Evaluation Harness

TODO describe the fixed SmolVLA policy, observation/action interface, LIBERO or ManiSkill environment, seeds, action budgets, and logging artifacts.

B. Candidate Chunk Generation

TODO define stochastic chunk sampling from the fixed lightweight VLA. Specify candidate count, sampling parameters, action horizon, and whether candidates share the same current observation and subgoal.

TABLE I

POLICY-ONLY BASELINE REPRODUCTION ON LIBERO. FILL WITH FINAL PAPER-SCALE NUMBERS ONLY.

Condition	Goal	Object	Spatial	Long	Avg
Policy-only	TODO	TODO	TODO	TODO	TODO
Reference	TODO	TODO	TODO	TODO	TODO

C. Imagined Outcome Rendering

TODO define the first proof path: clone simulator state, roll out each candidate chunk without committing it to the real evaluation environment, and render the resulting outcome image. Explain that this is an oracle-imagination upper-bound diagnostic. Later work can replace this with a learned action-conditioned visual dynamics or video prediction model.

D. VLM-Guided Chunk Selection

TODO define the visual judge inputs: current observation, predicted outcome image, candidate metadata, and current subgoal. The judge ranks candidates before execution. Only the selected chunk is committed to the benchmark environment.

E. Success Semantics

TODO state explicitly: final task success is the environment benchmark success signal, e.g., `info["success"]` or `is_success`. The VLM judge is a pre-execution candidate selector, not a final success detector.

IV. EXPERIMENTS

A. Setup

TODO list suites, task IDs, episodes per task, compute environment, simulator versions, policy checkpoint, and action-step protocol.

B. Policy-Only Baseline

TODO summarize the current routed SmolVLA baseline and cite the local artifact paths in the camera-ready or supplement, not in the blinded submission.

C. Imagine-Then-Act Selection Results

TODO compare `policy_only`, random candidate selection, oracle imagined candidate selection, and VLM-guided imagined candidate selection. Report success, action steps, wall-clock latency, candidate count, and judge accuracy against environment outcomes where available.

TABLE II
IMAGINED CHUNK SELECTION ABLATION. KEEP ALL CONDITIONS
MATCHED BY SEEDS, BASE POLICY, AND COMMITTED ACTION BUDGET
UNLESS EXPLICITLY STATED.

Condition	Episodes	Success	N	Latency
Policy-only	TODO	TODO	1	TODO
Random chunk	TODO	TODO	TODO	TODO
Oracle imagined selector	TODO	TODO	TODO	TODO
VLM imagined selector	TODO	TODO	TODO	TODO

V. DISCUSSION

TODO explain when imagined chunk selection helps, when it fails, and whether semantic visual judging is actually better than random or oracle-state heuristic selection.

VI. LIMITATIONS

TODO include limited task coverage, simulator-only evidence, privileged simulator-clone imagination, VLM judge brittleness, latency from evaluating multiple candidates, no weight updates, and no real robot claim unless real evidence is added.

VII. CONCLUSION

TODO one paragraph: what was tested, strongest measured result, and why pre-execution imagined chunk selection is a useful control layer for lightweight VLA policies.

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